FULL STACK DEVELOPER
AI & MACHINE LEARNING
DISTRIBUTED SYSTEMS

Mausam Kar

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Building scalable things for a more intelligent world.

ABOUT ME

Computer Science undergraduate at **VIT Bhopal** with a **9.13 CGPA**, building **full-stack, microservices and AI-powered** products across web and mobile. Currently a **Data Science Intern** at Prodigy Infotech, with a proven track record of shipping **RAG-based systems**, event-driven architectures supporting **10,000+ concurrent users**, and winning **1st place** at an Interclub Hackathon. Passionate about scalable systems, applied AI, and writing clean, production-ready code.

9.13	5	10K+	3	1st
CGPA VIT Bhopal	Independent Microservices	Concurrent Users Supported	Major Projects Built & Shipped	Interclub Hackathon

WORK EXPERIENCE

Prodigy Infotech — Data Science Intern

Present | Remote

Data Science • ML • Analytics

Working on end-to-end data science problems across analysis, modelling, and visualisation.

- Performed **end-to-end data analysis and preprocessing** including data cleaning, feature engineering, and exploratory data analysis to extract actionable insights from large datasets
- Implemented **machine learning models** using Python, NumPy, Pandas, and Scikit-learn for supervised and unsupervised learning problems including classification and clustering tasks
- Developed and evaluated **predictive models** using performance metrics such as accuracy, precision, and recall, and created data visualizations using Matplotlib and Seaborn to support data-driven business decisions

PROJECTS

Shikshak — Enterprise Learning Management System

Next.js • Node.js • Kafka • MongoDB • Redis • Docker

EdTech • Microservices

- **Architected microservices-based LMS** with event-driven design using Apache Kafka, enabling horizontal scaling and fault tolerance across 5 independent services (API Gateway, Auth, Courses, Payment, RAG)
- Implemented **service decomposition strategy** isolating compute-intensive operations (video processing, AI inference) into dedicated microservices for independent scaling based on workload
- Built **RAG-powered study assistant** with Redis vector database and LLM integration for semantic search and context-aware response generation across course materials
- Developed **real-time AI proctoring** using MediaPipe and TensorFlow.js for browser-based monitoring, and integrated Better-Auth for secure role-based access control

PROJECTS (CONTINUED)

CampusSync AI-Driven Educational Management Platform

React • Node.js • MongoDB • AWS

Serverless • AI

- Designed and developed an **AI-driven serverless educational ERP system** enabling real-time data synchronization, autonomous workflows, and personalized academic support for educational institutions
- Built a **context-aware AI assistant** for predictive student intervention, automated grading, and intelligent content generation using machine learning insights and natural language processing
- Architected a scalable event-driven system using serverless architecture supporting **over 10,000 concurrent users** with high availability and zero-operations deployment
- **Published project report** in International Journal and College Technical Journal

Real Scout Real Estate Mobile Application

React Native • Expo • TypeScript • Appwrite • NativeWind

Mobile • Real Estate

- Built **cross-platform real estate mobile app** using React Native and Expo with Google OAuth authentication and Appwrite backend for data management, authentication, and file storage
- Implemented **dynamic property listings** with advanced search and filtering by type, price range, and location with real-time data synchronization using Appwrite SDK
- Designed **scalable architecture** with clean separation of concerns, custom hooks for state management, and responsive UI using NativeWind (Tailwind CSS for React Native)

TECHNICAL SKILLS

Programming & Fundamentals
C++, C, Java, Python, TypeScript, JavaScript, Data Structures and Algorithms, Object-Oriented Programming

Web Technologies
React, Angular, Node.js, Express.js, HTML, CSS, Spring Boot

Mobile Development
React Native, Expo, NativeWind, Expo Router

Databases
MySQL, MongoDB, PostgreSQL, Prisma, Drizzle ORM, Appwrite

Tools & Technologies
Docker, Git, GitHub, RESTful APIs, Apache Kafka, Upstash Redis, WebSockets, Jest, Postman, Linux, AWS, Google Cloud Platform, Vercel, Render, Agile Methodology, CI/CD

Machine Learning
Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Google Gemini, Pinecone, RAG

EDUCATION

Bachelor of Technology (Computer Science and Engineering)
CGPA: 9.13 / 10.0
VIT Bhopal University | Sep 2024 – May 2028

Applied Machine Learning with Python
University of Michigan (Coursera), 2024

AWARDS & ACHIEVEMENTS

1st Position, Interclub Hackathon
Built a RAG-based AI system for context-aware information retrieval, 2024

Published CampusSync Report
International Journal and College Technical Journal, 2024

Modern Web Animations
Smooth scrolling and interactive transitions using GSAP and Lenis

LEADERSHIP & ACTIVITIES

Core Member
Linpack Club

Led 2 hackathons and organized 3 large-scale speaker sessions and technical workshops including fake currency detection using MATLAB, increasing member participation by 50 percent and new memberships by 30 percent